

Omais Hassan

Islamabad, Pakistan

+923105663511 | omaishasan17@gmail.com | LinkedIn

Seasoned software engineer with 5+ years of experience in full-stack web development, API design, workflow optimization, and emerging technologies. Proven expertise in building scalable microservices, improving frontend performance, and developing AI-powered chatbots. Adept at collaborating with cross-functional teams in Agile environments to deliver high-quality software solutions.

WORK EXPERIENCE

RIT Services GmbH

Senior Software Developer

Germany (Remote) | Nov 2022 - Present

- Improved page load times and responsiveness by identifying and addressing inefficiencies in React components, leading to increased user engagement.
- Engineered secure API infrastructure for model training and evaluation, incorporating encryption and access logging to mitigate data exposure risks and maintain data integrity.
- Contributed technical expertise to shape product roadmaps for 10+ full-stack web applications based on feasibility and scalability, reducing development costs by identifying technically infeasible features early in the product development lifecycle.
- Designed scalable microservice architecture implementing Repository, Dependency Injection, and MVC patterns, resulting in highly testable and maintainable codebase with clean separation of concerns
- Implemented horizontal scaling strategies for backend services, distributing traffic across multiple instances to maintain optimal performance under peak loads, resulting in enhanced system resilience to failures.
- Onboarded new developers by providing comprehensive training on team workflows, coding standards, and development environments, contributing to improved adherence to coding standards.

NextPak Agile Solutions

Full Stack Developer

Islamabad, Pakistan | Sep 2020 - Nov 2022

- Collaborated with cross-functional teams of UI/UX designers and product managers in Agile/Scrum framework to transform business requirements into technical specifications and user stories
- Deployed and managed applications across AWS EC2, Amplify, Lambda, and Fargate, as well as GCP Cloud Run, implementing scalable, reliable, and cost-efficient cloud environments with automated CI/CD and strong uptime performance.
- Developed backend for large-scale SaaS platforms using MySQL and Sequelize ORM with advanced database optimization including connection pooling, query caching, and horizontal scaling
- Managed Docker containerization and deployment pipelines, resulting in reduced downtime for critical application services.
- Wrote clean, modular, and testable code following best practices in version control (Git), code reviews, and continuous integration
- Conducted comprehensive front-end performance audits utilizing Lighthouse, resulting in optimized images and minimized render-blocking JavaScript to enhance user experience.

SKILLS

Languages & Frameworks: React, Node.js, TypeScript, Python, FastAPI, Express, Nest.js, Golang

Databases: PostgreSQL, MySQL, MongoDB, Firebase, Sequelize, SQLAlchemy

Cloud & Devops: Nginx, Docker, Google Cloud Platform, CI/CD Pipelines, Amazon EC2, AWS Lambda, Amazon RDS

Tools: Git, OBS Studio, Agile, Jira, Linux, Portainer

Ai & Infrastructure: RAG, LangGraph, Ollama, WebRTC, Mediasoup, Redis, RabbitMQ, Bull Queue

Testing: Jest, Cypress

PROJECTS

Kidsgo - Custom RAG Chatbot & AI Advertisement Engine

Nov 2024 - Present

Situation: Kidsgo, a high-traffic kids-courses platform with 10K daily visitors, needed smarter search, personalization, and better ad performance. Task: Improve content discovery and build an intelligent system that boosts user engagement and ad relevance. Action: Designed and built a custom RAG-based AI chatbot answering course and parenting queries in real time. Developed Europe's first AI-driven advertisement engine tailored for kids-course providers. Integrated AI search and ads deeply with Kidsgo's back-office CMS, enabling live content updates and instant metadata sync. Optimized retrieval pipelines, embeddings, and indexing for high-volume content. Result: Delivered intelligent, personalized discovery that increased search relevance and user interaction across the site. Impact: Improved user satisfaction and significantly enhanced monetization capability on a 10K/day organic platform.

GroundSync — Aviation & Logistics Real-Time Operations Platform

Aug 2024 - Present

Situation: Airports, airlines, and logistics operators lacked a shared real-time system for ground operations. Task: Build a transparent, multi-stakeholder collaboration platform with real-time events, integrations, and data consistency. Action: Developed using NestJS, ReactJS, and PostgreSQL, supported by microservices and event-driven architecture. Implemented modules for flights, tasks, SLAs, turnaround operations, real-time alerts, and geo-task tracking. Integrated with external airport systems (Stuttgart, Frankfurt and Vienna) and airlines (Ryanair and Lufthansa Cargo). Designed dashboards for airlines, handlers, and airport managers. Result: Unified platform offering complete operational visibility and improved coordination. Impact: Reduced delays, minimized disputes, and improved operational efficiency across aviation stakeholders.

Redezeit (Lower Saxony Parliament) — Session Management & Streaming

Mar 2022 - Present

Situation: The parliament needed a digital tool to manage speaker times, agendas, and live session streaming. Task: Build a full-stack platform integrating session logic with real-time recording. Action: Built modules for speaker timing, party duration allocation, agenda management, and live updates. Integrated with OBS for real-time streaming/recording automation. Added role-based control dashboards for admins and moderators. Result: A reliable, modern application replacing manual tracking and stream handling. Impact: Improved transparency, session control accuracy, and overall broadcast quality for a public institution.

Loqum AI — AI-Driven Digital Healthcare Platform

Nov 2024 - Dec 2025

Situation: Healthcare workflows were fragmented, with no structured pre-consultation screening and disconnected pharmacy, lab, and radiology services. Task: Build an end-to-end digital healthcare platform enabling AI-assisted patient screening, doctor decision-making, and integrated medical service fulfillment. Action: Led backend development for Patient, Doctor, and ICC/Dispensary modules, implementing AI-based initial patient screening that shared structured insights with doctors, enabling prescriptions for tests and medicines. Built a centralized ICC panel to manage labs, clinical procedures, radiology, and medicine orders with doorstep delivery, supported by secure, scalable APIs and role-based access control. Result: Delivered a unified platform connecting patients, doctors, and healthcare service providers in a single workflow. Impact: Reduced consultation turnaround time, improved diagnostic efficiency through AI pre-screening, and enabled reliable end-to-end healthcare delivery.

NPuLS — Cloud-Native Survey/Feedback Platform

May 2023 - Nov 2023

Situation: The platform needed modern UI/UX and scalable backend infrastructure to handle varying user activity. Task: Build a reliable, serverless application capable of scaling automatically. Action: Developed using Node.js (backend) and React.js (frontend). Deployed on Google Cloud Platform using serverless functions, Firestore, and automatic scaling patterns. Designed optimized APIs for low-latency responses and improved user experience. Implemented authentication, permissions, feedback forms, analytics pages, and event tracking. Result: Stable, scalable platform requiring minimal maintenance. Impact: Improved reliability and ensured consistent performance under unpredictable user loads.

Travos - Travel & Hospitality SaaS Platform

Feb 2022 - Jan 2023

Situation: Travel agencies, hotels, and sellers struggled with fragmented systems for reservations, pricing, and operational management.Task: Create an end-to-end SaaS platform unifying booking workflows, pricing logic, and seller dashboards.Action:Built the platform using Node.js, Vue.js, and a microservices architecture supporting multiple sellers/agencies.Implemented modules for reservations, dynamic pricing, operations, and reporting.Designed secure multi-tenant architecture enabling hotel chains and agencies to manage data independently.Integrated calendars, payment workflows, availability engines, and role-based dashboards.Result: Delivered a scalable SaaS solution enabling multiple travel partners to manage day-to-day operations from a unified platform.Impact: Significantly reduced operational overhead and improved organizational efficiency for agencies and hotels.

DashCafe - E-commerce + Subscription + Campaign Platform

May 2021 - Oct 2022

Situation: DashCafe needed a single platform combining store, subscriptions, fundraising, and content publishing.Task: Deliver a modern, scalable full-stack solution.Action:Built the backend using NestJS, frontend with Next.js, and database on AWS RDS.Deployed on AWS ECS and Amplify, ensuring scalability and optimized delivery.Developed modules for online ordering, coffee subscription renewals, fundraising campaigns, and blog publishing.Integrated payments, admin dashboards, customer accounts, and automated emails.Result: A polished, high-performing platform managing sales, recurring revenue, and customer engagement.Impact: Enabled brand growth, increased sales channels, and provided a professional digital presence.

EDUCATION

FAST NUCES

Islamabad, Pakistan

B.Sc. - Electrical Engineering (Computer Systems)

Aug 2017 - Jul 2021

TRAINING / COURSES

AI & LLM Engineering Mastery: GenAI, RAG Complete Guide

Clean Code with Java: Learn Simple Design, Refactoring & TDD

Privacy and Data Protection, Certified GDPR Compliance